

$$\frac{\partial z}{\partial x} = n x^{n-1} \phi\left(\frac{y}{x}\right) - x^n \phi'\left(\frac{y}{x}\right) \left(\frac{y}{x^2}\right) = n x^{n-1} \phi\left(\frac{y}{x}\right) - y x^{n-2} \phi'\left(\frac{y}{x}\right)$$

$$= x^{n-1} \left[n \phi\left(\frac{y}{x}\right) - \frac{y}{x} \phi'\left(\frac{y}{x}\right) \right] = x^{n-1} \cdot \phi'\left(\frac{y}{x}\right).$$

Thus, $\frac{\partial z}{\partial x}$ is a homogeneous function of degree $(n - 1)$.

Property3: Let $z = f(x, y)$ be a homogeneous function of degree n . Then we have $xf_x + yf_y = nf(x, y)$. This result is known as Euler's theorem.

Check Your Progress 2

- 1) Find the total differential of the function $y = \sin x_1 + \cos x_2$.
.....
.....
.....
.....
- 2) The function $q = AL^\alpha K^{1-\alpha}$, $A, \alpha > 0$ is known as the Cobb-Douglas production function. q , L and K are the level of output, labour and capital respectively.
Prove the following:
 - a) Cobb-Douglas production function is homogeneous of degree one.
 - b) The marginal and average productivities of the two inputs depend on the ratio of the two inputs only.
 - c) Elasticity of substitution is unity.
 - d) Iso-quants are downward sloping and convex to the origin.
.....
.....
.....

11.5 LET US SUM UP

This unit provided us with the understanding of the concept of partial derivatives, when we deal with functions of several variables. We learnt about the higher order derivatives and partial derivatives after gaining an understanding of basic partial derivatives. Then we moved on and discussed about total differential and total derivative. In the latter half of the unit, we learnt the technique for differentiation of multivariate cases, and noted how the concept is applied in economics. The rules to be kept in mind while dealing with implicit functions were also outlined. Lastly, we have applied the notion of partial differentiation to illustrate one important class of functions, namely homogeneous functions and we also discussed properties of such functions.

11.6 KEY WORDS

- Homogeneous Function** : A function which satisfies $f(tx, ty) = t^n(x, y)$.
- Partial Derivative** : Partial derivative refers to the derivative of a multivariate function when only one of the independent variable is allowed to change, other variables remaining constant; e.g., the function $U = f(x_1, x_2)$,
- Total Derivative** : Through total derivative, we measure the rate of change of the dependent variable owing to any change in a variable on which it depends, when none of the variables is assumed to be constant.
- Total Differential** : Rate of change of a variable in terms of another variable, without assuming that all other variables are held constant.

11.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) $Q_L = LK^2$, $Q_K = L^2K$.
- 2) $f_x = 1 = f_y$, Thus, $f_{xx} = f_{yy} = 0$
 $f(x, y) = y \sin x^2 + x^3 \cos y$. Find f_{xx} , f_{yy} .
- 3) $f_x = 2yx \cos x^2 + 3x^2 \cos y$
 $f_y = \sin x^2 - x^3 \sin y$
 $f_{xx} = 2y \cos x^2 - 4x^2 y \sin x^2 + 6x \cos y$
 $f_{yy} = -x^3 \cos y$
- 4) By Young's theorem, $f_{yx} = f_{xy}$, hence, $f_{xy} = f_{yx} = 4e^{2x+2y}$.

Check Your Progress 2

- 1) $dy = \cos x_1 dx_1 - \sin x_2 dx_2$.
- 2) Hint: Proceed along the lines for the proof of Property 2.
- 3) The function $q = AL^\alpha K^{1-\alpha}$, $A, \alpha > 0$ is known as the Cobb-Douglas production function. q , L and K are the level of output, labour and capital respectively.

The equation of the iso-quant is given by $\bar{q} = AL^\alpha K^{1-\alpha}$. Take that $\bar{q}$ is fixed level of output for different possible combinations of L and K . The slope is given by $\frac{d^2k}{dL^2} > 0$. Thus, the iso-quant is convex.

11.8 EXERCISES

Q1. Suppose that the cost function for a manufacturer is given by

$$C(x) = (10^{-6})x^3 - .003x^2 + 5x + 1000$$

a) Describe the behaviour of the marginal cost.

b) Sketch the graph of $C(x)$.

Ans. a) The first two derivatives of $C(x)$ are given by

$$C'(x) = (3 \cdot 10^{-6})x^2 - .006x + 5$$

$$C''(x) = (6 \cdot 10^{-6})x - .006.$$

The marginal cost function $y = (3 \cdot 10^{-6})x^2 - .006x + 5$ has its graph as a parabola that opens upward. Since $y = C''(x) = .000006(x - 1000)$, we see that the parabola has a horizontal tangent at $x = 1000$.

So, the minimum value of $C'(x)$ occurs at $x = 1000$. The corresponding y -coordinate is $(3 \cdot 10^{-6})(1000)^2 - .006 \cdot (1000) + 5 = 3 - 6 + 5 = 2$.

Consequently, at first, the marginal cost decreases. It reaches a minimum of 2 at production level 1000 and increases thereafter. Now make the graph on your own.

b) Try making the graph with the help of this information. We know that $C'(x)$ is never zero, so there are no relative extreme points. Since $C'(x)$ is always positive, $C(x)$ is always increasing (as any cost curve should). Moreover, since $C'(x)$ decreases for x less than 1000 and increases for x greater than 1000, we see that $C(x)$ is concave down for x less than 1000, is concave up for x greater than 1000, and has an inflexion point at $x = 1000$. Keep in mind that the inflexion point of $C(x)$ occurs at the value of x for which marginal cost is a minimum.

Q2. The demand equation for a certain product is $p = 6 - 1/2x$ rupees. Find the level of production that results in maximum revenue.

Ans. In this case, the revenue function $R(x)$ is

$$R(x) = x \cdot p = x(6 - 1/2 x)$$

$$= 6x - 1/2 x^2 \text{ rupees.}$$

The marginal revenue is given by $R'(x) = 6 - x$. The graph of $R(x)$ is a parabola that opens downward. It has a horizontal tangent precisely at those x for which $R'(x) = 0$ —that is, for those x at which marginal revenue is 0. The only such x is $x = 6$. The corresponding value of revenue is

$$R(6) = 6 \cdot 6 - 1/2 (6)^2 = 18 \text{ rupees}$$

Thus, the rate of production resulting in maximum revenue is $x = 6$, which results in total revenue of 18 rupees.

UNIT 12 DIFFERENTIAL CALCULUS: FUNCTIONS OF SEVERAL VARIABLES

Structure

- 12.0 Objectives
- 12.1 Introduction
- 12.2 Indefinite Integrals
 - 12.2.1 The Integral Sign
- 12.3 Properties of Indefinite Integrals
 - 12.3.1 Integral of a Power of x
 - 12.3.2 Method of Substitution
- 12.4 Integration of Trigonometric Functions
- 12.5 Integration by Parts
- 12.6 Some Useful Formulae
- 12.7 Determination of the Constant of Integration
- 12.8 Definite Integrals
 - 12.8.1 Finding Area
 - 12.8.2 Definition of Definite Integral
 - 12.8.3 Properties of Definite Integral
- 12.9 Fundamental Theorem of Calculus
 - 12.9.1 Evaluation of Definite Integrals
- 12.10 Integration with Chain Rule
 - 12.10.1 Integration with Chain Rule and Substitution
- 12.11 Improper Integrals
 - 12.11.1 Convergent and Divergent Integrals
- 12.12 Integration with Partial Fractions
- 12.13 Let Us Sum Up
- 12.14 Key Words
- 12.15 Answers/Hints to Check Your Progress Exercises
- 12.16 Exercises

12.0 OBJECTIVES

After going through this unit, you should be able to:

- explain integration as a mathematical process for problem solving ;
- perform integration of various types of functions using different integration methods: substitution rule, integration by parts, trigonometric substitution, partial fractions, rational substitution 0 ; and
- evaluate improper integral sand use the see valuation processes and definitions to compute an improper integral if it is convergent or divergent.

12.1 INTRODUCTION

This unit is devoted to finding and computing integrals. There are two types of integrals that we will be looking at in this unit, viz., Indefinite Integrals and Definite Integrals. The first half of the unit is devoted to indefinite integrals, properties of indefinite integrals, integrals of trigonometric functions. The latter half is devoted to definite integrals wherein we will understand about properties of definite integrals, fundamental theorem of calculus, integration with chain rule, improper integrals, integration with partial fractions. In order to equip you with application of these concepts of integration in terms of solving problems, some economic applications of definite and indefinite integration are also discussed in the unit.

12.2 INDEFINITE INTEGRALS

Indefinite integral is reverse of the concept of derivative. The differential calculus is used to derive the rate of change (or, derivative) of a given function whereas indefinite integration reverses such a process and finds the unknown function whose rate of change (derivative) is given. Thus, differentiation is just like finding square root of a number while integration is the process of finding the square.

The integral of function $f(x)$ is written symbolically as $\int f(x)dx$, which is read as “*the integral of $f(x)$ with respect to x* ” and $f(x)$ is called the ‘*integrand*’. We can write $\int f(x)dx = F(x)$, provided $F(x)$ is the derivative of $f(x)$ i.e., $f(x) = dF(x)/dx$. It is immediately clear that the function $F(x)$ is not unique, because if $F(x)$ has the derivative $f(x)$, then so has the function $F(x) + C$ where C is any arbitrary constant. That is why the value of the indefinite integral is always written by adding an **arbitrary constant C** as $\int f(x)dx = F(x) + C$ and then **C is called ‘the constant of integration’**. As we wish to perform the **opposite process** to differentiation, it is called “**anti-differentiation**” and we will call it “**integration**”. Let us look at few basic examples.

Example: If we know that $dy/dx = 3x^2$, and need to know the function from which this derivative has come from, then we “undo” the differentiation process. It is easy to see that $y = x^3$ is one anti-derivative of $dy/dx = 3x^2$.

Example 12.1 : Find $\int (x^2 - 5)dx$

As we know from our knowledge of differentiation that

$$d(x^3/3 - 5x + C)/dx = (x^2 - 5)$$

so, we can write

$$\int (x^2 - 5)dx = x^3/3 - 5x + C$$

In general, if $\int f(x)dx = F(x) + C$, $\int f(x)dx = F(x) + C$, then we say $F(x) + C$ is **indefinite integral** of $\int f(x)dx$.

12.2.1 The Integral Sign

The sign, viz $\int$, is an elongated “S” which stands for “sum”. The sum-symbol for integration is used in view of the fact, which we will see later, that the integral is the sum of the areas of infinitesimally thin rectangles. Sometimes we may write a capital letter to signify integration. For examples, we write $F(x)$ to mean the integral of $f(x)$. In that usage we have: $F(x) = \int f(x)dx$.

12.3 PROPERTIES OF THE INDEFINITE INTEGRALS

Let us now proceed to the properties of the integrals, most of which are derived from the corresponding properties of derivatives.

- 1) a) $\int k f(x)dx = k \int f(x)dx$, where k is any number. So, we can factor multiplicative constants out of indefinite integrals.
- b) $\int x^n dx = x^{n+1}/(n+1)$

Proof: a) Suppose that $F(x)$ is an anti-derivative of $F'(x) = f(x)$. Then by the basic properties of derivatives we also get $(kF(x))' = kF'(x) = k f(x)$.

So $kF(x)$ is an anti-derivative of $k f(x)$; i.e., $(kF(x))' = k f(x)$.

Thus, $\int k f(x)dx = k \int f(x)dx + C = k \int f(x)dx$. Similarly,

- b) Integral of a power: As $d(x^{n+1}/(n+1))/dx = x^n$, therefore $\int x^n dx = x^{n+1}/(n+1)$.

Example 12.2 : $\int 5x^2 dx = 5 \int x^2 dx = 5 \left(\frac{x^3}{3} + c \right)$

Ans.

Example 12.3 : Integrate $\int x^5 dx$

Ans. Using the formula obtain: $\int x^5 dx = \frac{x^{5+1}}{5+1} + C = \frac{x^6}{6} + C$

Example 12.4 : Integrate $\int 8x^6 dx$

Ans. $8 \frac{x^{6+1}}{6+1} = (8/7)x^7 + C$

See that 8 is a constant, so it is kept out in the beginning while calculating.

$$2) \quad \int f(x) \pm g(x) dx = \int f(x) dx \pm \int g(x) dx$$

Proof: Suppose that $F(x)$ is an anti-derivative of $f(x)$ and that $G(x)$ is an anti-derivative of $g(x)$. So, we have that $F'(x) = f(x)$ and $G'(x) = g(x)$. Basic properties of derivatives also tell us that

$(F(x) \pm G(x))' = (F'(x) \pm G'(x)) = f(x) \pm g(x)$. So $F(x) + G(x)$ is an anti-derivative of $f(x) + g(x)$ and $F(x) - G(x)$ is an anti-derivative of $f(x) - g(x)$. In other words,

$$\int f(x) \pm g(x) dx = F(x) \pm G(x) + C = \int f(x) dx \pm \int g(x) dx.$$

Example: 12.5

$$\int (4x^4 + 7x + 3) dx = \int (4x^4) dx + \int (7x) dx + \int 3 dx = \frac{4}{5}x^5 + \frac{7}{2}x^2 + 3x + c$$

Ans.

Example 12.6 : Integrate $dy = (5x^2 - 4x + 3) dx$

Ans.: Write $\int dy = \int (5x^2 - 4x + 3) dx$.

In the left side $\int dy = \int (1) dy = y$. Integration of the constant 1 with respect to y yields y

On the right-hand side, we integrate each of the terms, one at a time:

$$\int (5x^2 - 4x + 3) dx = 5 \frac{x^3}{3} - 4 \frac{x^2}{2} + 3x + C$$

So, putting it all together, we have the solution:

$$y = (5/3x^3)/3 - 2x^2 + 3x + C$$

Example 12.7: $\int \left(3x^2 + \sqrt{x} - \frac{5}{x^3} \right) dx$

Ans. Re-write the exponents to make it easier to integrate. So,

$$\int \left(3x^2 + \sqrt{x} - \frac{5}{x^3} \right) dx =$$

$$\int (3x^2 + x^{1/2} - 5x^{-3}) dx = x^3 + \frac{x^{3/2}}{\frac{3}{2}} - 5 \frac{x^{-2}}{-2} + C = x^3 + \frac{2x^{3/2}}{3} + 5 \frac{x^{-2}}{2} + C$$

Remarks: The value of the constant of integration can be uniquely determined by other constraints, e. g.

Example 12.8 : A particular curve's derivative is given by $\frac{dy}{dx} = 3x^2 - 2x$. The curve passes through the point (2,5). Find the equation of the curve.

Ans: The first step for this problem is to integrate the expression (i.e., find the anti-derivative). This will give us the expression for y : $\int (3x^2 - 2x) dx = x^3 - x^2 + C$

So, we have $y = x^3 - x^2 + C$

This represents a family of curves, and depends on the value of C for the y -intercept. We must now find the value of C from the information given in the question. Since the curve passes through $(2, 5)$, we substitute this value into $y = x^3 - x^2 + C$ to give: $5 = (2)^3 - (2)^2 + C$ or, $5 = 8 - 4 + C$.

So, $C = 1$ giving the required curve is $y = x^3 - x^2 + 1$. The graph of the curve is presented in Fig. 12.1.

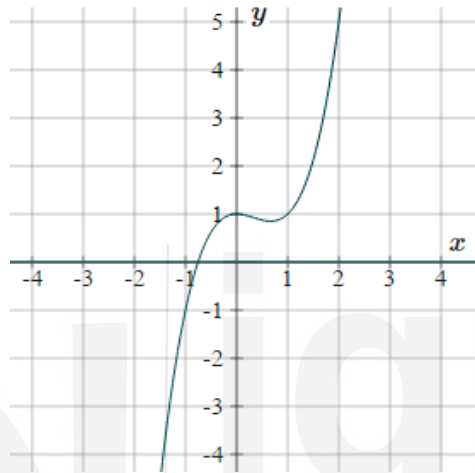


Fig. 12.1: Graph of the curve $y = x^3 - x^2 + 1$

Notice the curve passes through the point $(2, 5)$.

Example 12.9: Consider the integration $\int (2x^4 - 5)^6 x^3 dx$

An examination of the expression indicates that we cannot integrate it using the rules we have learned so far. Finding integrals of products of functions is generally tricky. In some cases, the method of substitution, discussed next, may be helpful.

If $\int u^n du = \frac{u^{n+1}}{n+1} + C$, it is true if $n \neq -1$. We need a substitution step, where $u(x)$ is some function of x .

Now let us go back to example 9 to see how to apply this formula.

Integrate $(2x^4 - 5)^6 x^3$ with respect to x , we use, as a starting point, the substitution $u = 2x^4 - 5$, because $2x^4 - 5$ is the expression in brackets in the question. Differentiating u gives: $du = 8x^3 dx$

On dividing both sides by 8, we get $1/8 du = x^3 dx$.

Rewriting the integration as $\int (2x^4 - 5)^6 x^3 dx = \frac{1}{8} \int u^6 du$, we integrate with respect to u we get $\frac{1}{8} \times \frac{u^7}{7} + C = \frac{u^7}{56} + C$.

Substituting back the value of x , you get $[(2x^4 - 5)^7 / 56] + C$

The method of substitution explains how a product in the integrand may be reduced to a single function: derivative of one of the factors in the product may subsume the other factor, as derivative of $(2x^4 - 5)$ subsumed x^3 , the other factor.

12.3.1 Method of Substitution

If $F(x) = \int f(x) dx$, then the indefinite integral can be obtained by resorting to transformation. If we take $x = g(y)$ then

$$\int f(x) dx = \int f[g(y)]g'(y) dy$$

See that $x = g(y) \Rightarrow dx = g'(y) dy$, $x = g(y) \Rightarrow dx = g'(y) dy$ where $g'(y)$ denotes derivative of $g(y)$ with respect to y .

If $\phi(y) = \int f[g(y)]g'(y) dy$, then

$F(x) = \phi[g^{-1}(.)]$ is the inverse of $g(.)$.

Example 12.10: Solve $F(x) = \int (1 + 5x)^{1/2} dx$

Let $y = 1 + 5x$

So, $dy = 5dx$ or $dx = 1/5 dy$

$$\begin{aligned} \therefore F(x) &= \frac{1}{5} \int y^{1/2} dy = \frac{1}{5} \cdot \frac{2}{3} y^{3/2} + c \\ &= \frac{2}{15} y^{3/2} + c = \frac{2}{15} (1 + 5x)^{3/2} + c \end{aligned}$$

Example 12.11: Solve $F(x) = \int \frac{dx}{(3x-1)^2}$

Let $y = 3x - 1$

So, $dy = 3dx$ or $dx = 1/3 dy$

$$\therefore F(x) = \frac{1}{3} \int \frac{dy}{y^2} = \frac{1}{3} \left(-\frac{1}{y} \right) + c = \frac{1}{3(3x-1)} + c$$

Example 12.12 : Evaluate $\int e^{2x} dx$, using substitution.

Let $u = 2x$ so $du = 2dx$ and $dx = 1/2 du$. Thus, we get $\frac{1}{2} \int e^u du$. Integration

gives us $\frac{1}{2} \int e^u du = \frac{1}{2} e^u = \frac{e^{2x}}{2}$.

Example 12.13: Find $\int 2x\sqrt{x^2+1} dx$

Here we let $g(x) = x^2 + 1$. We have $\int 2x\sqrt{x^2+1} dx = \int \sqrt{g(x)} g'(x) dx$

Now, we let $u = x^2 + 1$ which gives us

$du/dx = 2x$ which in turn gives us

$du = 2x dx$. Therefore, we have

$$\int 2x(\sqrt{x^2 + 1}) dx = \int \sqrt{u} du = \frac{2u^{3/2}}{3} + C.$$

Check Your Progress 1

- 1) What do you understand by indefinite integral?

.....

- 2) Find $\int 4 dx$.

.....

- 3) Find $\int (x^2 + 3) dx$.

.....

- 4) Find $\int 9e^{-3x} dx$.

.....

- 5) Find $\int \frac{1}{x+2} dx$.

.....

12.4 INTEGRATION OF TRIGONOMETRIC FUNCTIONS

Let us recall the basic trigonometric identities before understanding their integration.

- i) Ratio Trigonometric identities:

$$\tan A = \sin A / \cos A$$

$$\cot A = 1 / \tan A = \cos A / \sin A$$

- ii) Reciprocal Trigonometric Identities:

$$\sin A = 1 / \operatorname{cosec} A \text{ or } \operatorname{cosec} A = 1 / \sin A$$

$$\cos A = 1 / \sec A \text{ or } \sec A = 1 / \cos A$$

$$\tan A = 1 / \cot A$$